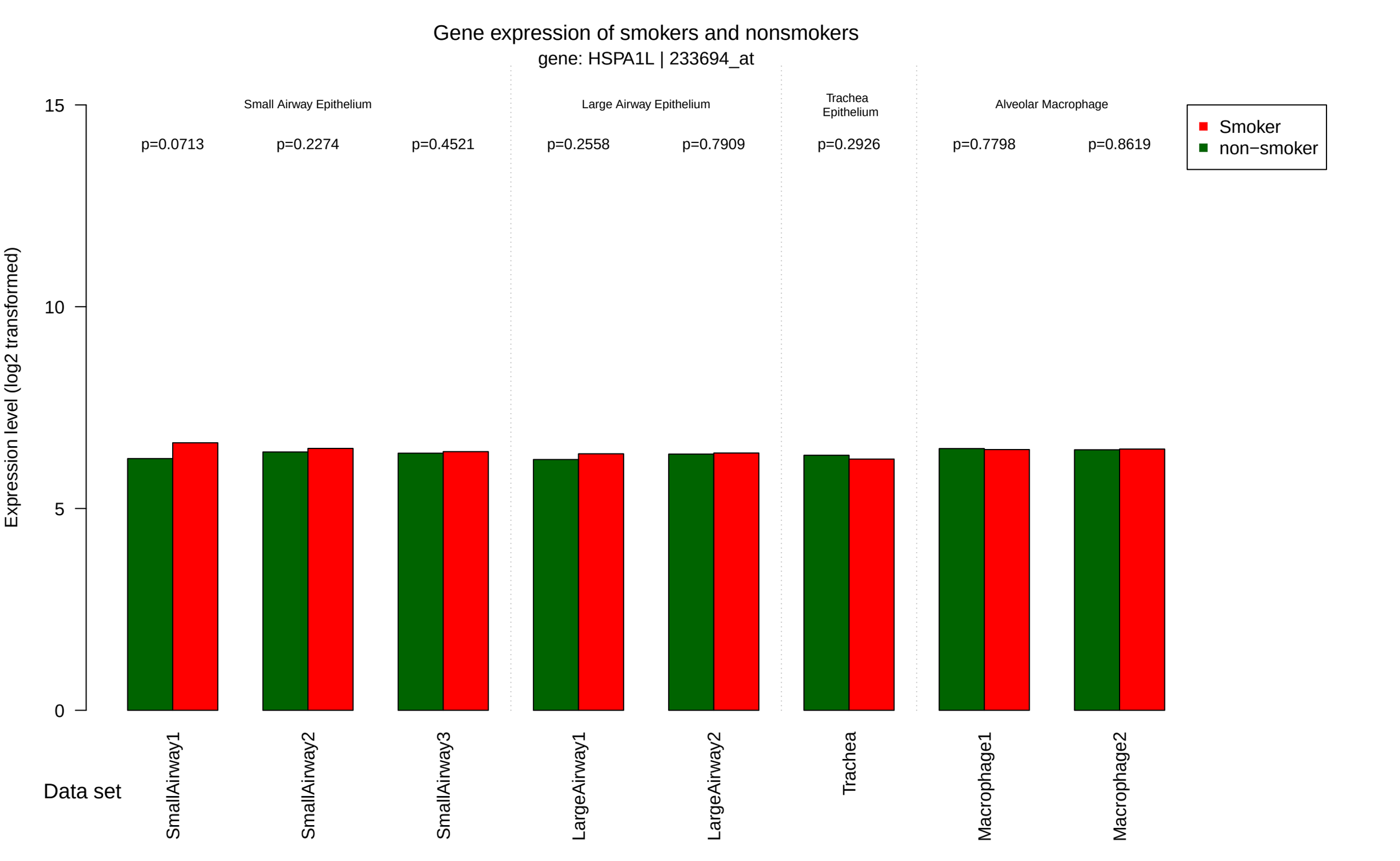
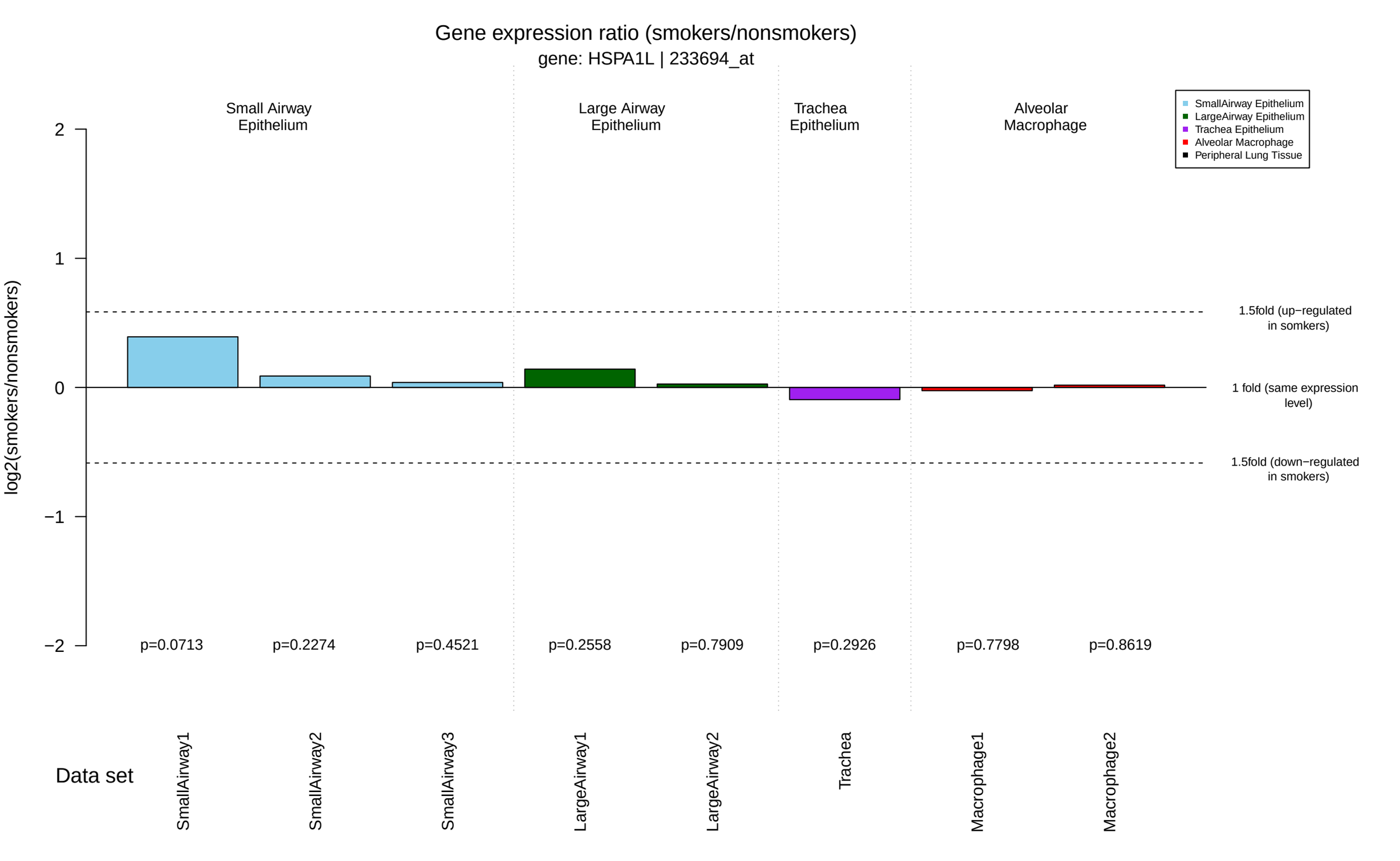
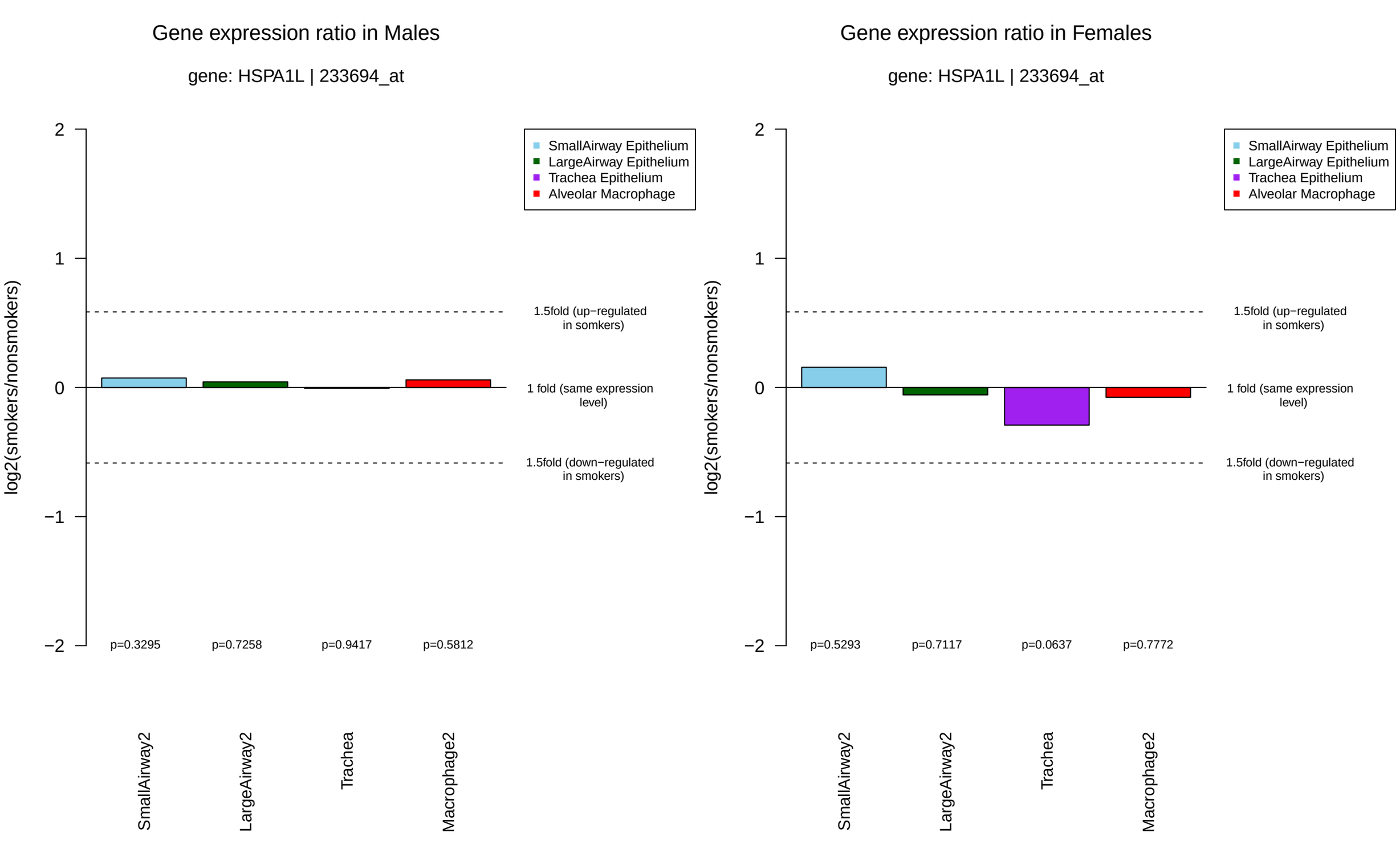
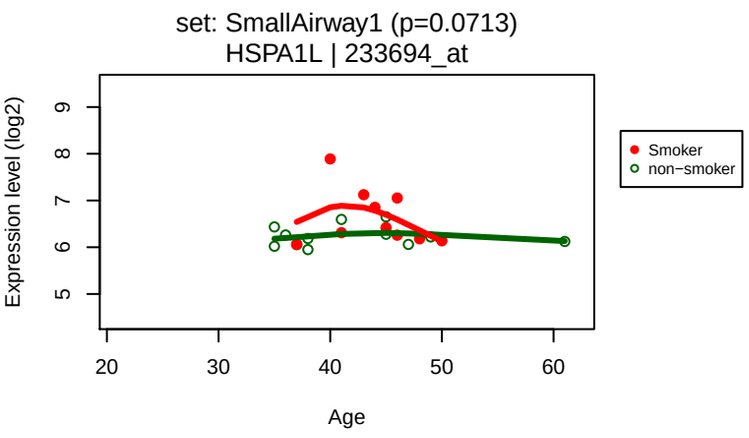
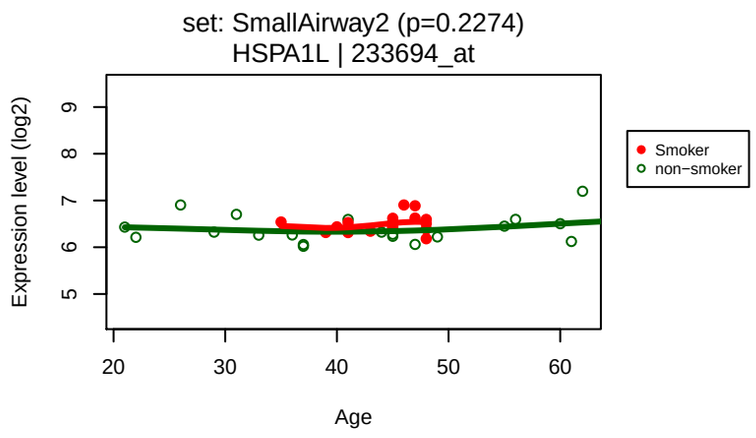
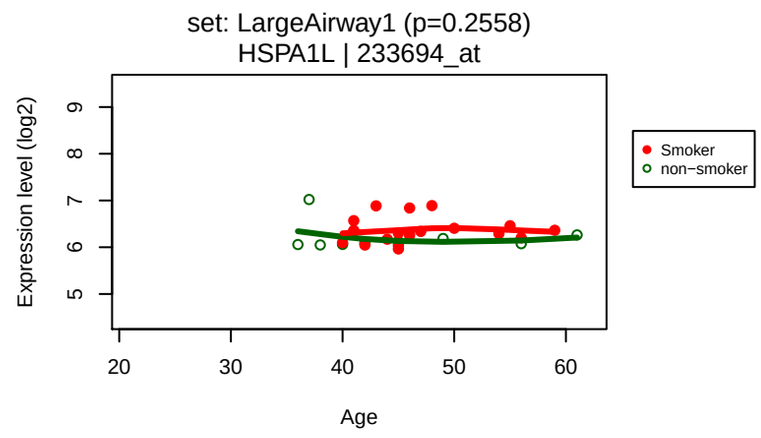

Gene expression of smokers and nonsmokers
gene: HSPA1L | 233694_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0713
p=0.2274
p=0.4521
p=0.2558
p=0.7909
p=0.2926
p=0.7798
p=0.8619
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

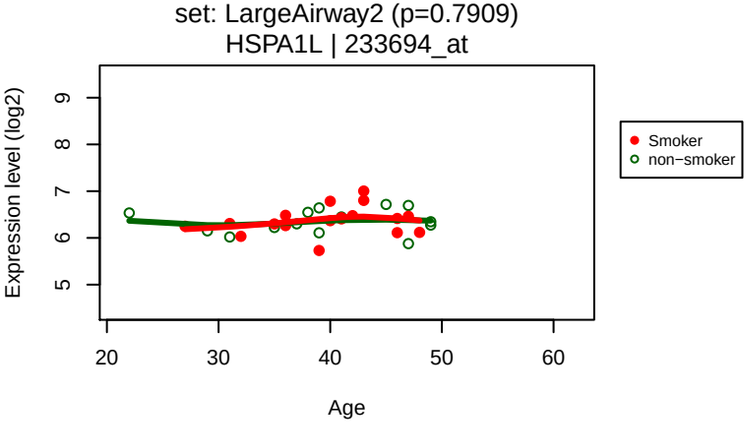

Gene expression ratio (smokers/nonsmokers)
gene: HSPA1L | 233694_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0713
p=0.2274
p=0.4521
p=0.2558
p=0.7909
p=0.2926
p=0.7798
p=0.8619
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

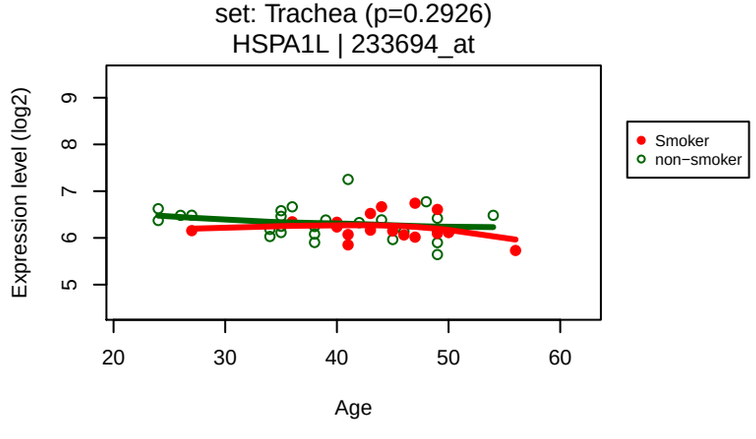

Gene expression ratio in Males
gene: HSPA1L | 233694_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3295
p=0.7258
p=0.9417
p=0.5812
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA1L | 233694_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5293
p=0.7117
p=0.0637
p=0.7772
SmallAirway2
LargeAirway2
Trachea
Macrophage2

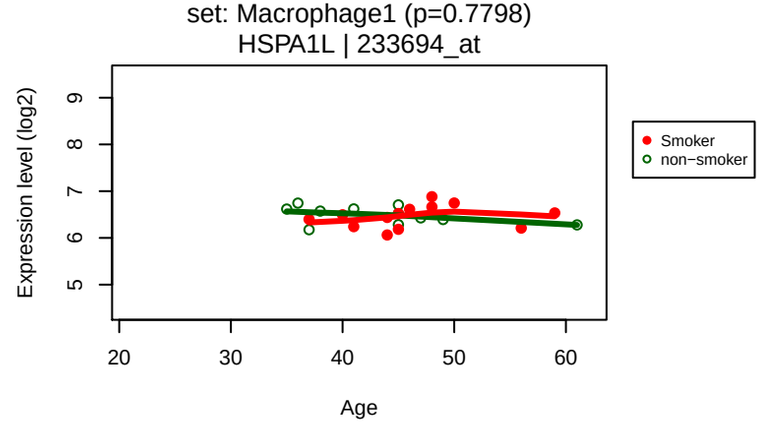

set: SmallAirway1 (p=0.0713)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker

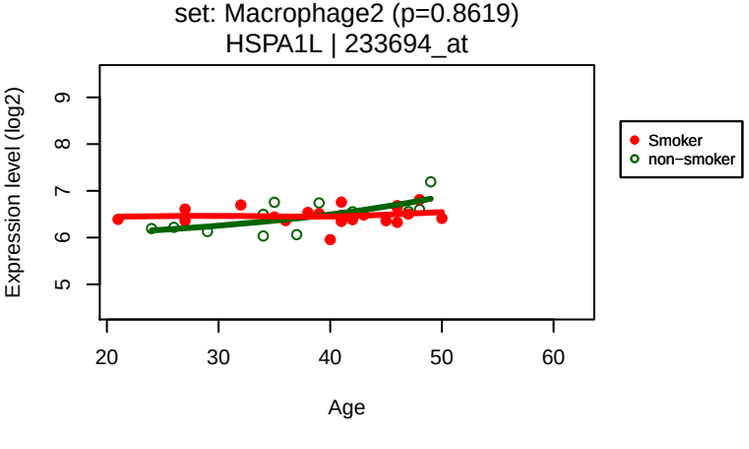

set: SmallAirway2 (p=0.2274)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker

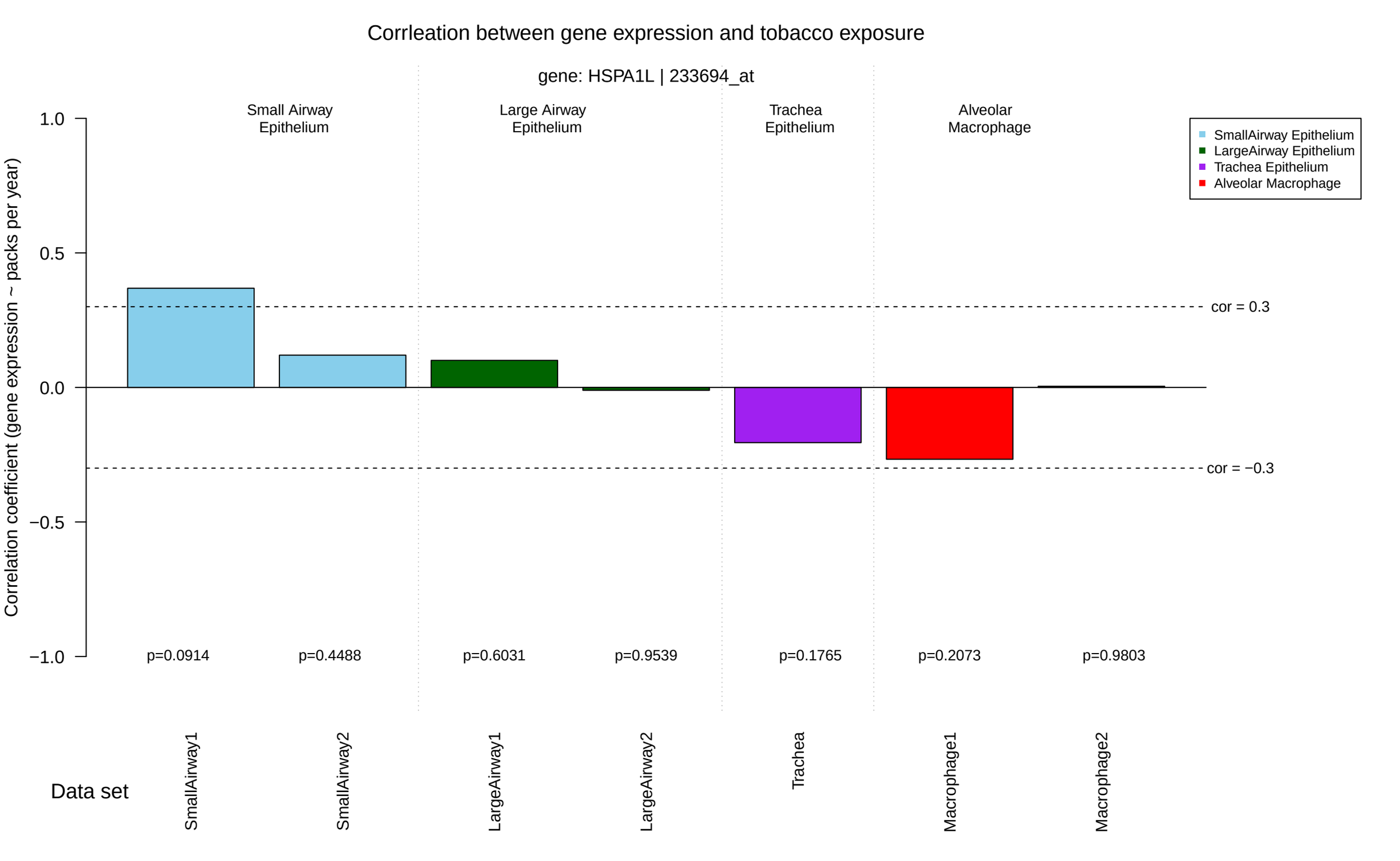

set: LargeAirway1 (p=0.2558)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7909)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2926)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7798)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8619)
HSPA1L | 233694_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA1L | 233694_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0914
p=0.4488
p=0.6031
p=0.9539
p=0.1765
p=0.2073
p=0.9803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2